

MPAS-URBAN QUICKSTART | GUIDE 1 OF 3

Install HKUST-MPAS

Start from the official HKUST-MPAS source, compile both atmospheric executables, and add the MPAS-Urban geographical data.

OUTCOME You will have a standard HKUST-MPAS installation for learning and new experiments, plus a clearly separated optional source tree for the HRAIN2025 NTDK cutoff experiment.

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Before you begin

This guide supplements rather than replaces the official MPAS-Atmosphere tutorial. New users should first complete the official tutorial or Quick Start Guide and confirm that a standard MPAS case can be compiled and run.

- Official tutorial: <https://mpas-dev.github.io/atmosphere/tutorial.html>
- Quick Start Guide:
https://www2.mmm.ucar.edu/projects/mpas/site/documentation/users_guide/quick_start.html

Use a Unix/Linux server. On an HPC system, prefer the compiler, MPI, NetCDF, and PnetCDF modules supported by the site. Keep one compiler/MPI family for MPAS and every MPI-linked library.

If the server does not yet provide a working MPAS build environment, first read the dependency sections of the official tutorial and then use Appendix A in this guide. Appendix A gives one complete user-space GNU/Open MPI example for a machine without an existing MPAS environment. On a managed HPC cluster, site-provided modules remain the preferred route.

Step 1 | Check the existing MPAS build environment

COMMAND

```
command -v gcc
command -v gfortran
command -v mpicc
command -v mpif90
command -v pnetcdf-config

mpicc --showme:command 2>/dev/null || mpicc -show
mpif90 --showme:command 2>/dev/null || mpif90 -show
```

CHECK For the gfortran build target, the MPI wrappers should resolve to a compatible GNU compiler family. Stop and correct the module environment if the wrappers point to a different compiler family.

Step 2 | Download the standard HKUST-MPAS source

Use the HKUST-MPAS team repository and its `hkust-dev` branch as the default source for MPAS-Urban learning and new independent experiments.

STANDARD SOURCE

```
mkdir -p "$HOME/MPAS"
cd "$HOME/MPAS"

git clone --depth 1 --branch hkust-dev --single-branch https://github.com/HKUST-MPAS/HKUST-MPAS.git

cd HKUST-MPAS
git branch --show-current
test -f Makefile
test -d src/core_atmosphere
```

EXPECTED `git branch --show-current` prints `hkust-dev`.

Step 3 | Optional HRAIN2025 NTDK cutoff source

Skip this step for a standard MPAS-Urban installation. The following repository is a case-author derivative used for the HRAIN2025 cutoff experiment; it is not the official HKUST-MPAS main branch and does not represent the default HKUST-MPAS physics configuration.

CASE-AUTHOR DERIVATIVE

```
cd "$HOME/MPAS"

git clone --depth 1 --branch ntdk-grid-cutoff --single-branch https://github.com/Liuzh223/HKUST-MPAS-Official.git HKUST-MPAS-NTDK

cd HKUST-MPAS-NTDK
git branch --show-current
```

NAME NOTE The word `Official` is part of this GitHub repository name only. The `ntdk-grid-cutoff` branch is maintained for this case. There is no separate terminal command named `cutoff`.

OPTION NOTE This derivative adds `config_cu_disable_dx`. The value is selected later in Guide 3; it is not a compile command.

Step 4 | Compile both MPAS executables

Set `PNETCDF` to the prefix already prepared through the official tutorial or your cluster modules. Leave `PIO` unset to use the bundled `SMIOL` path in this MPAS version.

BUILD ENVIRONMENT

```
# Example only: change this to the PnetCDF prefix on your system.
export PNETCDF="$HOME/opt/mpas-libs"
unset PIO

pnetcdf-config --version
```

Choose exactly one source directory before compiling:

```
# Standard HKUST-MPAS (default)
cd "$HOME/MPAS/HKUST-MPAS"

# HRAIN2025 cutoff derivative (optional; use instead of the line above)
# cd "$HOME/MPAS/HKUST-MPAS-NTDK"
```

BUILD

```
make -j 4 gfortran CORE=init_atmosphere PRECISION=single
make clean CORE=atmosphere
make -j 1 gfortran CORE=atmosphere PRECISION=single
```

VERIFY

```
test -x init_atmosphere_model
test -x atmosphere_model
ls -lh init_atmosphere_model atmosphere_model
```

BUILD NOTE The serial atmosphere build avoids Fortran module races observed on some systems. Increase the parallel build count only after the toolchain has been verified.

Step 5 | Download the standard MPAS static data

DOWNLOAD

```
mkdir -p "$HOME/MPAS_DATA"
cd "$HOME/MPAS_DATA"

curl -fL -O https://www2.mmm.ucar.edu/projects/mpas/mpas_static.tar.bz2

tar -xjf mpas_static.tar.bz2
test -d "$HOME/MPAS_DATA/mpas_static"
```

Use the absolute path to `$HOME/MPAS_DATA/mpas_static` as `config_geog_data_path` during initialization.

Step 6 | Add the MPAS-Urban CGLC-LCZ data

DOWNLOAD AND EXTRACT

```
cd "$HOME/MPAS_DATA"

curl -fL -O https://github.com/HKUST-MPAS/HKUST-MPAS/releases/download/LCZ_dataset_for_MPAS_Urban/mpas_cglc_lcz.tar.gz

mkdir -p "$HOME/MPAS_DATA/mpas_static/mpas_cglc_lcz"
tar -xzf mpas_cglc_lcz.tar.gz -C "$HOME/MPAS_DATA/mpas_static/mpas_cglc_lcz"

test -s "$HOME/MPAS_DATA/mpas_static/mpas_cglc_lcz/index"
```

The HKUST-MPAS source already supplies the MPAS-Urban parameter tables. Do not download replacement urban tables unless a specific code version instructs you to do so.

Ready for Guide 2

- `init_atmosphere_model` and `atmosphere_model` exist.
- The standard MPAS static dataset is available.
- `mpas_static/mpas_cglc_lcz/index` exists.

Appendix A | Build the MPAS dependencies from scratch

Use this appendix only when the server has no compatible MPI, NetCDF, or PnetCDF environment. Install all user-built libraries under one prefix and use the same GNU compiler and Open MPI wrappers for every MPI-linked package.

A.1 | Example versions

- Open MPI 4.1.6
- zlib 1.3.1
- HDF5 1.14.6
- NetCDF-C 4.9.3
- NetCDF-Fortran 4.6.4
- PnetCDF 1.12.3

These are pinned tutorial examples, not a requirement to use the newest release. Changing several versions at once makes failures harder to diagnose.

A.2 | Create the local prefix

```
export MPAS_PREFIX="$HOME/opt/mpas-libs"
export MPAS_LIB_SRC="$HOME/src/mpas-libs"
mkdir -p "$MPAS_PREFIX" "$MPAS_LIB_SRC"

export PATH="$MPAS_PREFIX/bin:$PATH"
export LD_LIBRARY_PATH="$MPAS_PREFIX/lib:$MPAS_PREFIX/lib64"
export CPATH="$MPAS_PREFIX/include"
export LIBRARY_PATH="$MPAS_PREFIX/lib:$MPAS_PREFIX/lib64"
export PKG_CONFIG_PATH="$MPAS_PREFIX/lib/pkgconfig:$MPAS_PREFIX/lib64/pkgconfig"
export CPPFLAGS="-I$MPAS_PREFIX/include"
export LDFLAGS="-L$MPAS_PREFIX/lib -L$MPAS_PREFIX/lib64"
```

A.3 | Optional user-space Open MPI

Skip this section when compatible mpicc, mpif90, and mpiexec commands already exist.

```
export MPI_PREFIX="$HOME/opt/openmpi-4.1.6"
mkdir -p "$HOME/src" "$MPI_PREFIX"
cd "$HOME/src"

curl -fL -O https://download.open-mpi.org/release/open-mpi/v4.1/openmpi-4.1.6.tar.gz
tar -xzf openmpi-4.1.6.tar.gz
cd openmpi-4.1.6

./configure --prefix="$MPI_PREFIX" CC=gcc CXX=g++ FC=gfortran
make -j 8
make install

export PATH="$MPI_PREFIX/bin:$PATH"
export LD_LIBRARY_PATH="$MPI_PREFIX/lib:$LD_LIBRARY_PATH"
mpicc --showme:command
mpif90 --showme:command
```

A.4 | Build zlib

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://zlib.net/fossils/zlib-1.3.1.tar.gz
tar -xzf zlib-1.3.1.tar.gz
cd zlib-1.3.1
./configure --prefix="$MPAS_PREFIX"
make -j 8
make check
make install
```

A.5 | Build HDF5

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://github.com/HDFGroup/hdf5/releases/download/hdf5_1.14.6/hdf5-1.14.6.tar.gz
tar -xzf hdf5-1.14.6.tar.gz
cd hdf5-1.14.6
CC=gcc ./configure --prefix="$MPAS_PREFIX" --with-zlib="$MPAS_PREFIX" --enable-shared --
disable-fortran
make -j 8
make check
make install
```

A.6 | Build NetCDF-C

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://downloads.unidata.ucar.edu/netcdf-c/4.9.3/netcdf-c-4.9.3.tar.gz
tar -xzf netcdf-c-4.9.3.tar.gz
cd netcdf-c-4.9.3
CC=gcc ./configure --prefix="$MPAS_PREFIX" --disable-dap
make -j 8
make check
make install
nc-config --version
nc-config --has-nc4
```

A.7 | Build NetCDF-Fortran

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://downloads.unidata.ucar.edu/netcdf-fortran/4.6.4/netcdf-fortran-4.6.4.tar.gz
tar -xzf netcdf-fortran-4.6.4.tar.gz
cd netcdf-fortran-4.6.4
CC=gcc FC=gfortran ./configure --prefix="$MPAS_PREFIX"
make -j 8
make check
make install
nf-config --version
```

A.8 | Build PnetCDF with the selected MPI wrappers

```
cd "$MPAS_LIB_SRC"
curl -fL -O https://parallel-netcdf.github.io/Release/pnetcdf-1.12.3.tar.gz
tar -xzf pnetcdf-1.12.3.tar.gz
cd pnetcdf-1.12.3
CC=mpicc FC=mpif90 ./configure --prefix="$MPAS_PREFIX"
make -j 8
make check
make install
pnetcdf-config --version
```

A.9 | Export the MPAS build variables

```
export PATH="$MPAS_PREFIX/bin:$PATH"
export LD_LIBRARY_PATH="$MPAS_PREFIX/lib:$MPAS_PREFIX/lib64:$LD_LIBRARY_PATH"
unset NETCDF NETCDFF PNETCDF PIO
export PNETCDF="$MPAS_PREFIX"
```

```
which mpicc mpif90
pnetcdf-config --version
nc-config --version
nf-config --version
```

Return to Step 2 after these checks pass. If a make check command fails, do not continue to MPAS compilation until the compiler, MPI wrapper, header, and library paths are consistent.